

An Online Beauty Store Case Study: How to Measure Influencer Marketing Budget Through Chain Rule?

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Abstract

Digital skincare and beauty brands face a massive business hurdle: traditional digital ad auctions are becoming too expensive. As the Cost Per Click (CPC) rises, standard paid ads yield less impact, forcing businesses to find alternative ways to grow. This paper outlines how Amina Beauty Creation Ltd (ABC Ltd) solves this problem by using a simple mathematical tool called the calculus chain rule. Instead of relying on broken link-tracking clicks, ABC Ltd invests in rising and trendy influencers to achieve an efficient view harvest. We track how these top-of-funnel views spill over into high-intent marketplace searches on Shopee, Lazada, and TikTok Shop. This case study provides a step-by-step framework for data science teams to calculate the exact point of budget saturation, ensuring every marketing dollar spent contributes directly to net profit.

1 Introduction and Corporate Challenge

Amina Beauty Creation Ltd, known in the market as ABC Ltd, is an online-native beauty and skincare brand. The company sells its products entirely through e-commerce channels, specifically TikTok Shop, Shopee, and Lazada. Recently, ABC Ltd faced a critical growth blocker: the cost of running standard conversion ads on social media skyrocketed. As competitive bidding drove up the Cost Per Click (CPC), the business saw less financial impact from its daily ad spend.

To bypass this expensive ad auction system, the data science team at ABC Ltd shifted to a new strategy. Instead of paying for expensive clicks, they deployed their budget to secure an efficient view harvest by partnering with rising and trendy influencers. These creators are heavily favored

by modern interest-graph algorithms, allowing ABC Ltd to gain massive consumer attention at a fraction of the cost. However, because these viewers buy products later through organic marketplace searches rather than clicking direct tracking links, ABC Ltd needed a reliable way to justify this marketing budget to its stakeholders. The team solved this by turning the marketing funnel into a trackable mathematical equation.

2 The Mathematical Framework and Chain Rule Derivation

To find out exactly how spending money on trending video views translates into final sales, ABC Ltd models its online ecosystem as a chain reaction. Blended e-commerce revenue Y depends on marketplace intent signals I . These intent signals are generated by total video views V , which are purchased using the initial marketing spend X . By structuring these variables as a composite function, the data team applies the calculus chain rule to find the exact rate of change in revenue per dollar spent.

$$\frac{dY}{dX} = \frac{dY}{dI} \cdot \frac{dI}{dV} \cdot \frac{dV}{dX} \quad (1)$$

The first term, $\frac{dY}{dX}$, represents the Total Marginal Return on Investment. It measures the exact dollar value added to the company's total revenue for every single additional dollar injected into the influencer budget.

The second term, $\frac{dY}{dI}$, is the Marketplace Monetization Velocity. It tracks the average amount of gross revenue generated by each high-intent search or anchor link click once the customer arrives on a product page.

The third term, $\frac{dI}{dV}$, functions as the Platform Spillover Rate. It serves as a direct mathematical proxy for how well the video anchors the brand in the consumer's mind, calculating how many marketplace search actions are triggered per video view.

The final term, $\frac{dV}{dX}$, represents the View Harvesting Efficiency. It measures the total volume of video views the brand acquires per dollar spent, acting as the direct operational indicator of the media team's purchasing power.

3 Data Science Blueprint: Step-by-Step Data Gathering

To build this tracking system, the data science team extracts clean data directly from online platforms, bypassing the need for complex web cookies. The pipeline is executed through three systematic steps.

In the first step, the team gathers the baseline inputs of marketing spend X and total video views V . This data is collected weekly from the TikTok Seller Center analytics dashboard and cross-referenced with the monthly invoices provided by the partner influencer marketing agency.

In the second step, the team captures the mid-funnel intent signals I . To archive this, they download the weekly search volume data for the brand name keywords from the Shopee Seller Centre by navigating to the Business Insights portal under the Search Analysis tab. They repeat this exact process in the Lazada Business Insights panel to log matching brand search queries, and add the raw product anchor link clicks exported directly from the TikTok Shop backend.

In the third step, the team extracts the final revenue output Y . They download the gross completed order values across Shopee, Lazada, and TikTok Shop for identical timeframes, ensuring that all variables map cleanly to the exact same weekly calendar dates.

4 Data Processing and Baseline Isolation

Raw digital metrics cannot be thrown directly into the formula because they contain natural marketplace noise and time delays. ABC Ltd processes this data through two distinct steps.

First, the team isolates the true influencer lift by establishing an organic sales baseline. They measure the brand's natural search volume and revenue during quiet, post-campaign periods, such as the week immediately following a major double-day mega sale. This baseline is subtracted from active campaign periods to ensure only net new revenue is evaluated.

Second, to account for the fact that a customer might watch an influencer video on a Tuesday but wait until a weekend payday to purchase, the team applies a mathematical adstock transformation. This technique uses a decay factor to distribute video views across a multi-week window, aligning the traffic data perfectly with the delayed tail of marketplace purchases.

5 Hypothetical Application: The USD 5,000 Deployed Budget

To see how this framework functions in a real corporate scenario, let us look at a case where the data science team at ABC Ltd prepares to invest an initial budget of $X = \text{USD } 5,000$. The team must determine the exact view count threshold required to make this investment highly profitable and choose the correct tier of creators to deliver those numbers. If we assume the marketplace monetization velocity is evaluated at USD 2.00 per intent signal and the platform spillover rate sits at a stable 2.5 percent, the team can map out the required view harvesting efficiency using the chain rule. To

achieve an outstanding twelve-fold return on capital at this early stage of the growth curve, the total marginal return must equal twelve.

To determine the exact operational metrics required, the data science team runs the chain rule calculation step by step. First, they substitute the targets and fixed historical coefficients into the core differential equation:

$$12 = 2.00 \cdot 0.025 \cdot \frac{dV}{dX} \quad (2)$$

Next, the team simplifies the known product of the marketplace conversion rate and platform spillover velocity:

$$2.00 \cdot 0.025 = 0.05 \quad (3)$$

This simplifies the master equation, expressing the total target return as a direct function of the view acquisition velocity:

$$12 = 0.05 \cdot \frac{dV}{dX} \quad (4)$$

By running the calculation backward and isolating the final derivative term, the team solves for the required view harvesting efficiency:

$$\frac{dV}{dX} = \frac{12}{0.05} = 240 \text{ views per dollar spent} \quad (5)$$

Finally, knowing that the view harvesting efficiency represents the total volume of video views V divided by the total budget X , the team multiplies the required efficiency rate by the planned capital allocation of USD 5,000 to extract the absolute attention target:

$$V = 240 \cdot 5,000 = 1,200,000 \text{ total views} \quad (6)$$

To secure this target of 1,200,000 views without overspending, the team cannot rely on a single mega-influencer. A mega-influencer with over a million followers often charges the entire USD 5,000 budget for a single video post, but interest-graph algorithms frequently suppress commercial content from large accounts, resulting in an unpredictable and inefficient view harvest. Instead, the team must execute a micro-portfolio approach by contracting a cohort of eight to twelve rising and trendy micro-influencers who possess between 50,000 and 150,000 followers. Because these specific creators are currently favored by the platform algorithms, their content experiences strong natural distribution waves, allowing each video to easily pull in 100,000 to 150,000 views. By paying an average of USD 400 to USD 500 per creator across twelve accounts, the team successfully hits the 1,200,000 total view target while keeping the average cost per view capped at a highly efficient half a cent per view.

6 Story Behind the Visualisation

The core objective of this framework is to identify the precise boundary of profitable growth.

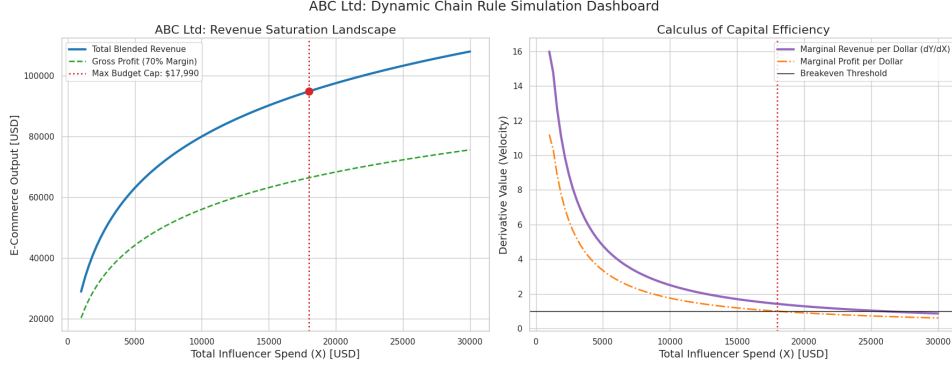


Figure 1: ABC Ltd: Dynamic Chain Rule Simulation Dashboard displaying the Revenue Saturation Landscape (left panel) and the Calculus of Capital Efficiency (right panel) to identify the maximum profitable budget ceiling at \$17,990.

The visualization is split into two distinct analytical dashboards that tell a continuous story of capital allocation. The left panel of Figure 1, titled “ABC Ltd: Revenue Saturation Landscape,” maps the total influencer budget against the company’s financial output. As the blue solid line shows, total blended revenue rises quickly at first but gradually bends horizontally. This curve demonstrates the law of diminishing returns; as the budget scales up, the brand exhausts the most relevant audiences, requiring more capital to reach new viewers. The green dashed line shows the 70% gross profit margin left after manufacturing and platform fees. The critical point on this chart is the red dot marked by the red dotted line, identifying the Max Budget Cap at exactly \$17,990. The right panel of Figure 1, titled “Calculus of Capital Efficiency,” explains exactly why \$17,990 is the mathematical ceiling. This graph plots the derivative velocities of the chain rule. The purple line shows the marginal revenue generated per dollar, while the orange dash-dot line displays the actual marginal profit returned to the business. The horizontal black line at 1.00 represents the absolute breakeven threshold, where one dollar spent returns exactly one dollar of profit. As ABC Ltd increases its budget, the efficiency of harvesting views drops, causing the orange profit velocity curve to slope downward. The red dotted line highlights the exact point where the orange curve crosses the breakeven line. Spending any amount beyond \$17,990 pushes the company into the danger zone, where the cost of acquiring new views outweighs the profit they bring in.